

Sentinel Exam: Decentralized AI-Proctored Examination System

**A MAJOR PROJECT SYNOPSIS SUBMITTED
IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF DEGREE OF**

**BACHELOR OF TECHNOLOGY
In
Artificial Intelligence & Machine Learning**

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2026

ABSTRACT

The rapid expansion of online education has significantly transformed the assessment landscape, but it has also introduced critical challenges in ensuring academic integrity, scalability, and trust. Conventional online examination systems rely heavily on centralized architectures, which are prone to data manipulation, single points of failure, and privacy concerns. These limitations undermine the credibility of remote assessments and create barriers to their widespread adoption. To address these issues, this project proposes a Decentralized AI-Proctored Examination System that integrates multi-modal artificial intelligence with blockchain-based auditability to create a secure and transparent examination environment.

The proposed system utilizes advanced AI-driven techniques to monitor candidate behavior in real time. It incorporates face recognition models such as FaceNet and ArcFace for continuous identity verification, ensuring that the registered candidate remains present throughout the examination. Additionally, object detection using YOLOv8 is employed to identify unauthorized objects such as mobile phones, books, or the presence of multiple individuals within the examination environment. To further enhance monitoring, gaze tracking is implemented using OpenCV and MediaPipe Face Mesh, enabling the system to analyze eye movements and detect deviations from the screen that may indicate suspicious behavior.

Complementing the visual analysis, the system integrates audio analytics through Voice Activity Detection (VAD) to identify anomalies such as multiple speakers or unusual background noise. These multi-modal inputs are combined within a behavior scoring model, which dynamically evaluates candidate actions using rule-based logic or machine learning techniques. The model assigns a risk score to each candidate, enabling real-time detection and flagging of suspicious activities during the examination.

A key feature of the system is its use of decentralized technologies to ensure data integrity and transparency. Critical examination logs and behavioral events are recorded on a blockchain network, making them immutable and tamper-proof. Large media files, including video recordings and snapshots, are stored using the InterPlanetary File System (IPFS), with their cryptographic hashes linked to the blockchain for secure verification. This approach eliminates reliance on centralized storage and enhances trust among stakeholders.

The system is designed using a microservices-based architecture, allowing modular integration of frontend interfaces, AI inference engines, and backend services. It supports real-time communication and scalable deployment, enabling efficient handling of large numbers of concurrent users. Furthermore, the solution is highly viable, requiring only standard hardware such as webcams and computers, and is optimized with lightweight AI models to function effectively under moderate network conditions.

Beyond its technical capabilities, the proposed system offers significant social impact by promoting fair and accessible online assessments. It reduces dependence on manual invigilation, minimizes malpractice, and provides a reliable solution for institutions in regions with limited physical infrastructure.

In conclusion, the Decentralized AI-Proctored Examination System presents a robust, scalable, and transparent framework that redefines digital examinations by combining intelligent monitoring with decentralized trust mechanisms, paving the way for next-generation education systems.